LAPORAN UJIAN TENGAH SEMESTER

Arsitektur Perancangan Perangkat Lunak

NAMA : SAMUEL ALEX BONAPARTE SIRAIT

NIM : 11423006

KELAS : 43TRPL1

INSTITUT TEKNOLOGI DEL

D-IV TEKNOLOGI REKAYASA PERANGKAT LUNAK

FAKULTAS VOKASI

Case Study 1 : MediTrack – Digital Healthcare Platform

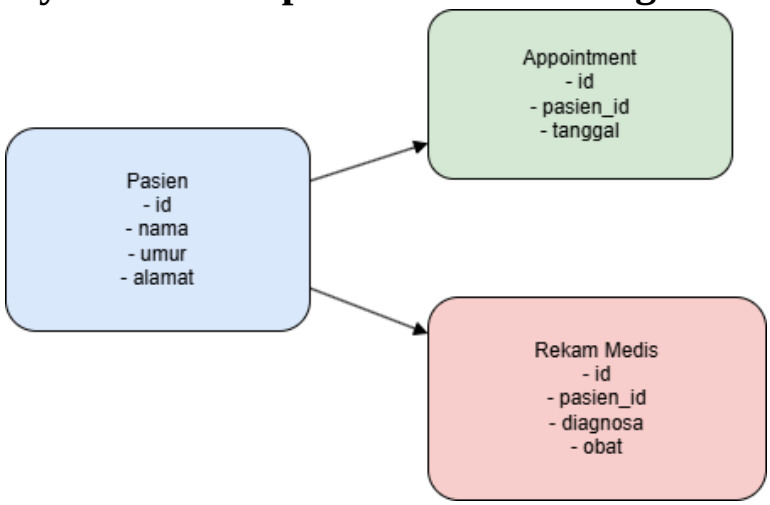
A. Design Thinking & Architecture Selection

Berdasarkan kebutuhan sistem MediTrack yang mencakup manajemen pasien, penjadwalan, dan rekam medis, sistem ini diimplementasikan menggunakan arsitektur Monolithic berbasis laravel (MVC).

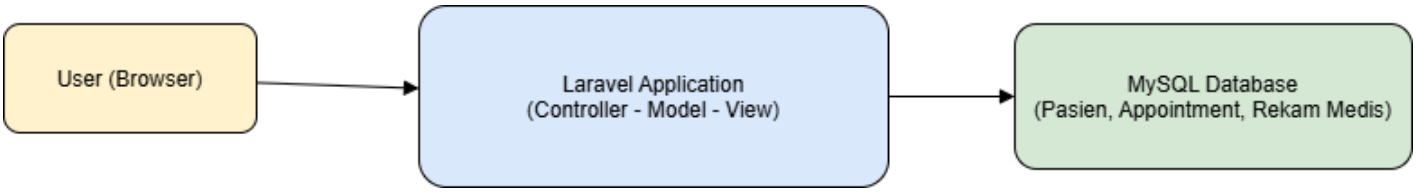
1. Justifikasi Pemilihan Arsitektur:
   Arsitektur monolith dipilih karena mampu memberikan kesederhanaan dalam pengembangan melalui pendekatan MVC pada Laravel, menjaga konsistensi data dalam satu basis data terpusat, serta mendukung maintainability melalui struktur kode yang terorganisir, extensibility dengan kemudahan penambahan fitur dalam satu codebase, dan scalability secara vertikal untuk sistem MediTrack dalam skala kecil hingga menengah.

2. Trade-offs dan Batasan:
   Meskipun mudah diimplementasikan, arsitektur monolith memiliki keterbatasan dalam hal skalabilitas, fleksibilitas integrasi dengan layanan eksternal, serta kompleksitas pemeliharaan ketika sistem berkembang karena seluruh modul terintegrasi dalam satu codebase.

B. System Decomposition & Modeling

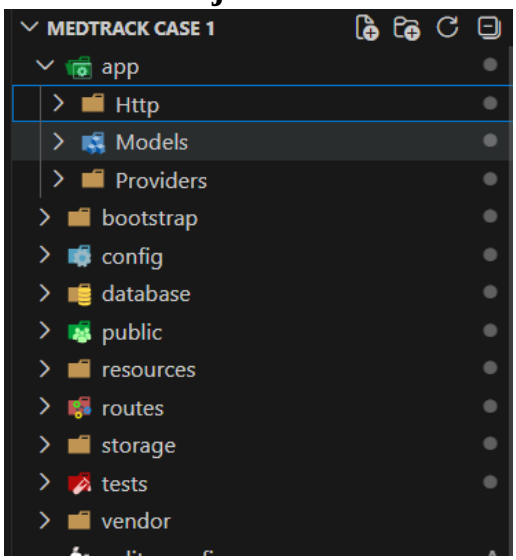


C. Architecture Visualization



D. Implementasi

- **Struktur Project**



- **Migration**

Pasien:

```
    Schema::create('pasien', function (Blueprint $table) {
$table->id();
$table->string('nama');
$table->integer('umur');
$table->text('alamat');
$table->timestamps();
```

Appointment:

```
{
    Schema::create('appointment', function (Blueprint $table) {
$table->id();
$table->foreignId('pasien_id')->constrained('pasien');
$table->date('tanggal');
$table->timestamps();
```

Rekam medis :

```
{
    Schema::create('rekam_medis', function (Blueprint $table) {
$table->id();
$table->foreignId('pasien_id')->constrained('pasien');
$table->text('diagnosa');
$table->text('obat');
$table->timestamps();
```

- **Model**

```
class Pasien extends Model
{
    protected $table = 'pasien';
    protected $fillable = ['nama','umur','alamat'];

    public function appointment(){
        return $this->hasMany(Appointment::class);
    }

    public function rekamMedis(){
        return $this->hasMany(RekamMedis::class);
    }
}
```

- **Controller**

```
    public function store(Request $request){
        Pasien::create($request->all());
        return redirect('/pasien');
    }
}
```

- **Routing**

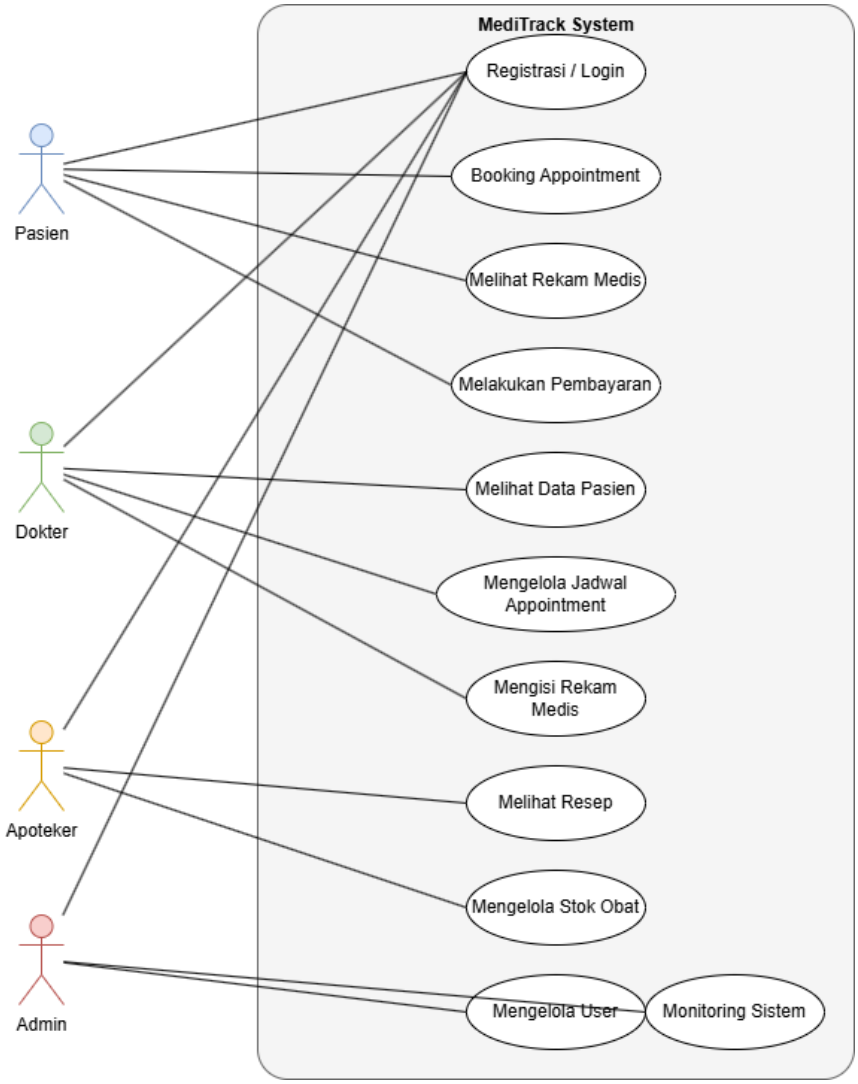
```
Route::get('/pasien', [PasienController::class, 'index']);
Route::get('/pasien/create', [PasienController::class, 'create']);
Route::post('/pasien/store', [PasienController::class, 'store']);
```

- **View**

```
@extends('layout')

@section('content')

<h2>Data Pasien</h2>

<a href="/pasien/create" class="btn btn-success mb-3">+ Tambah Pasien</a>

<table class="table table-bordered">
    <tr>
        <th>Nama</th>
        <th>Umur</th>
        <th>Alamat</th>
    </tr>

    @foreach($data as $p)
    <tr>
        <td>{{ $p->nama }}</td>
        <td>{{ $p->umur }}</td>
        <td>{{ $p->alamat }}</td>
    </tr>
    @endforeach
</table>
```

Case Study 2 : MediTrack – Digital Healthcare Platform

Provide the use case based on the case study!



Architecture Diagram